

Image Registration – 2

Feature-based image registration

Reference:

1. Matthew Brown and David G. Lowe, "Automatic panoramic image stitching using invariant features," International Journal of Computer Vision, 74, 1 (2007), pp. 59-73
<http://www.cs.ubc.ca/~lowe/papers/07brown.pdf>
2. CSE/EE486 Computer Vision I, Introduction to Computer Vision, CSE Department, Penn State University, Instructor: Robert Collins
<http://www.cse.psu.edu/~rtc12/CSE486/lecture15.pdf>

Image Stitching using Features



(a) Image 1

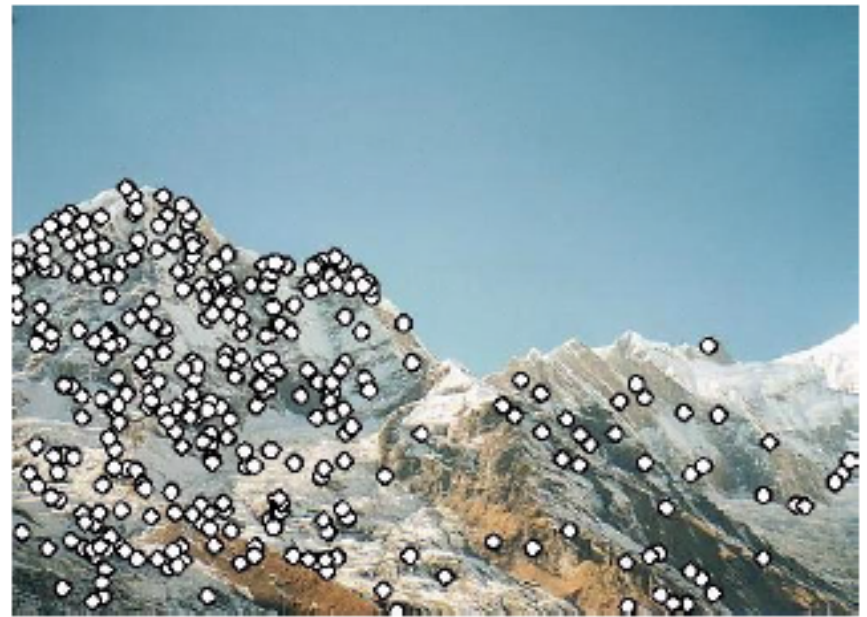


(b) Image 2

Image Stitching using Features



(c) SIFT matches 1



(d) SIFT matches 2

SIFT is the image feature which will be covered in class later

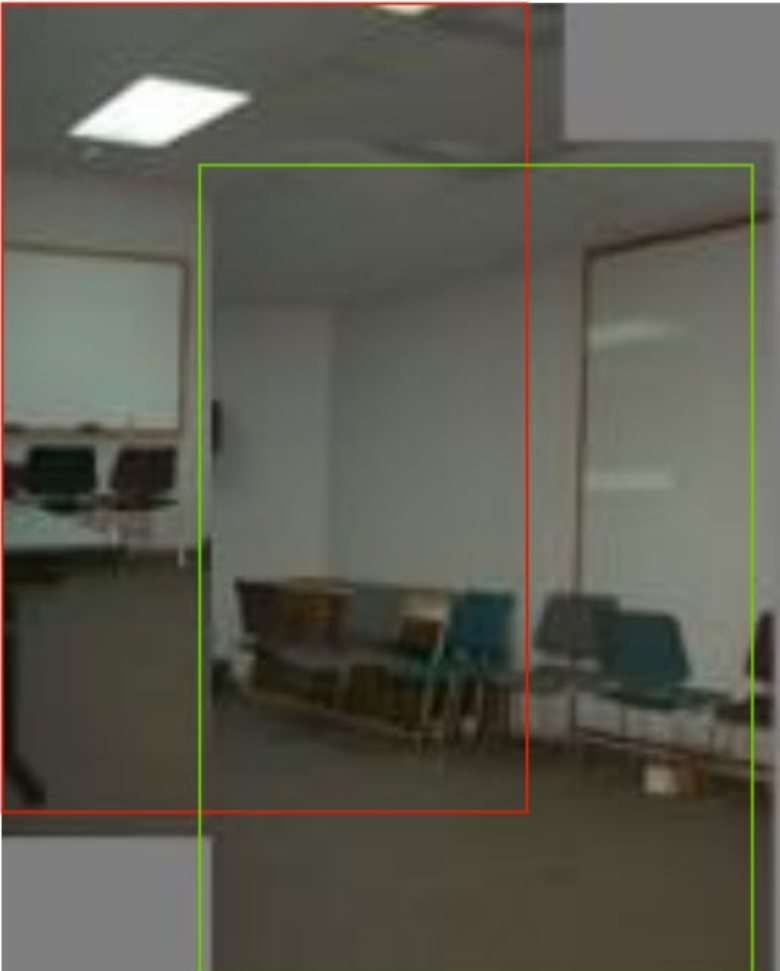
Image Stitching using Features



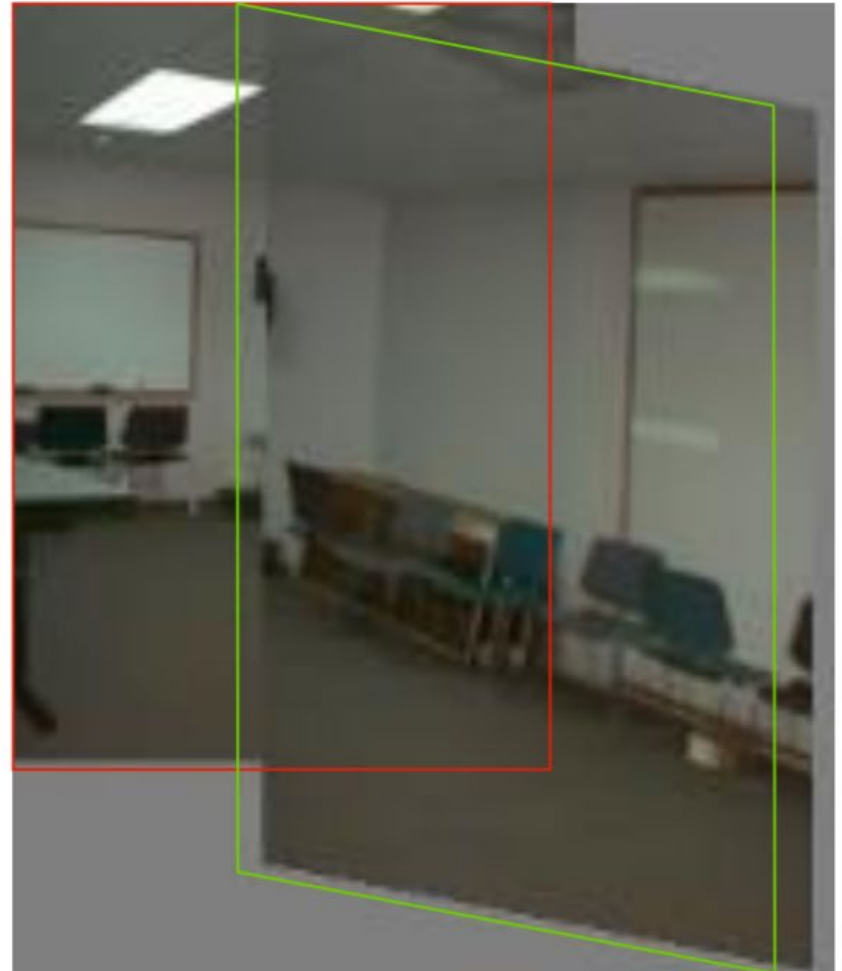
(g) Images aligned according to a homography

Registration with translation only and affine registration

translation



affine



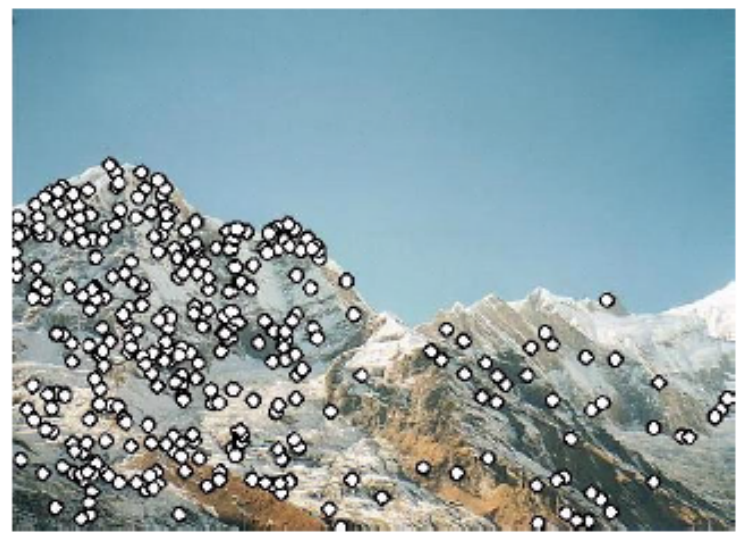
RANSAC

- How do we find the parameters of the affine registration when we have a collection of data points? Some points are valid (inliers), and some are invalid (outliers).
- View estimation as a two-stage process:
 - Classify data points as outliers or inliers
 - Fit model to inliers while ignoring outliers
- Technique: RANSAC – searches for random samples that lead to a fit in which many data points agree.
(RANdom SAmple Consensus)

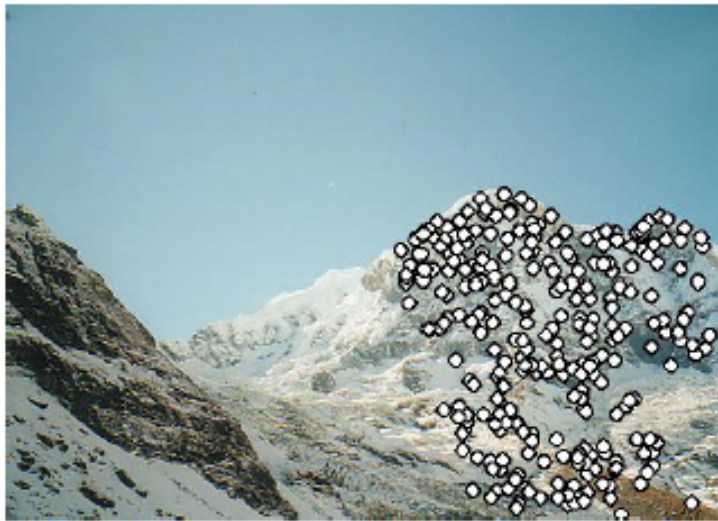
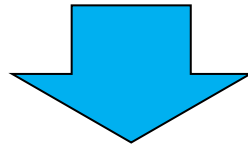
M. A. Fischler and R. C. Bolles (June 1981). "Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography". *Comm. of the ACM* **24**: 381--395.



(c) SIFT matches 1



(d) SIFT matches 2



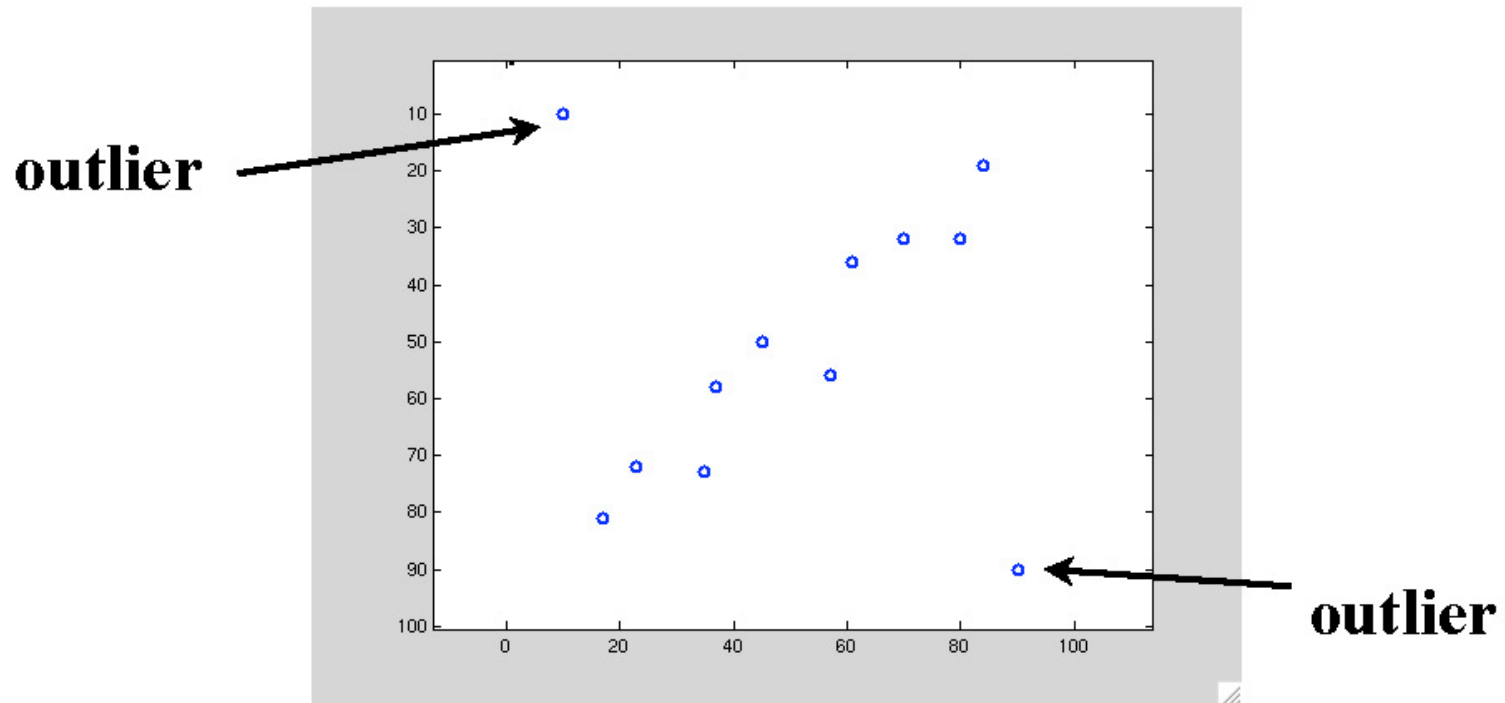
(e) RANSAC inliers 1



(f) RANSAC inliers 2

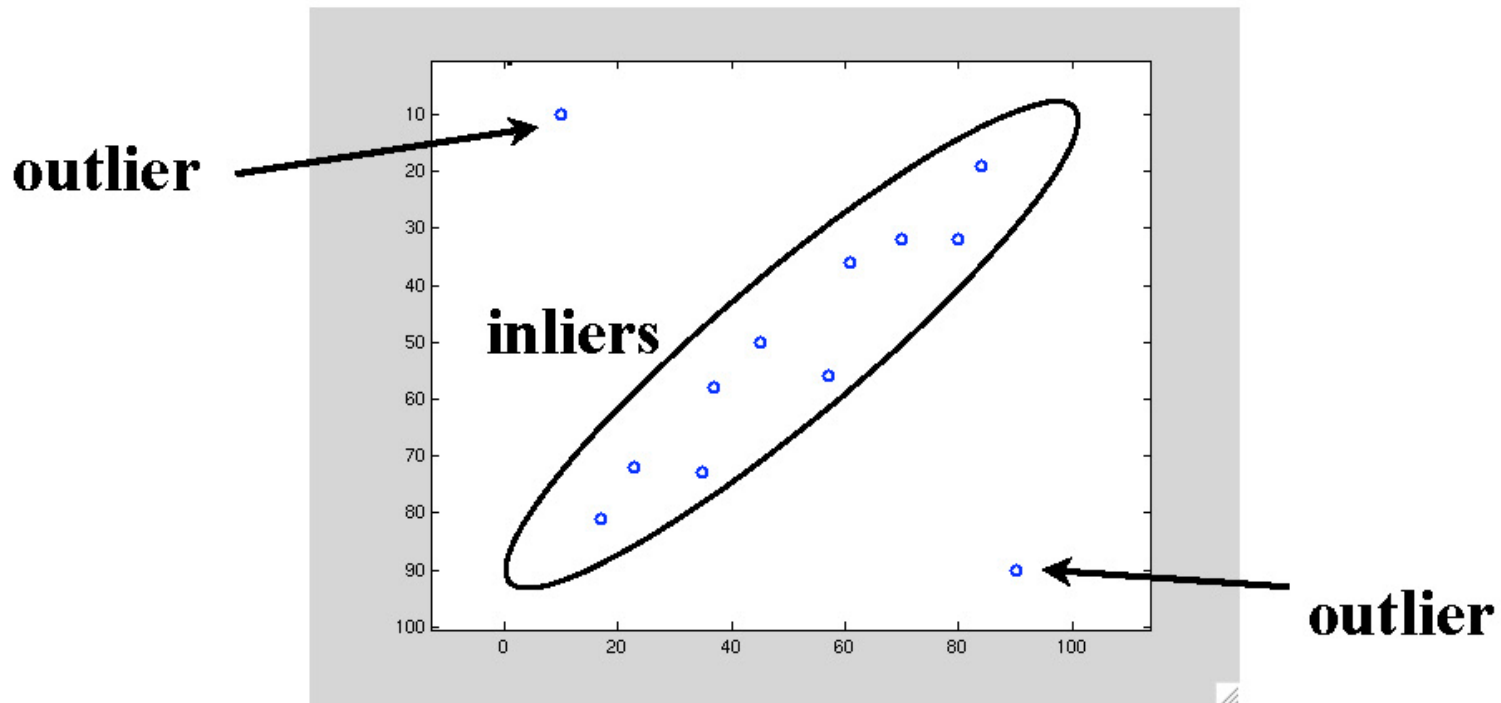
RANSAC

There are points (outliers) that are caused by measurement errors and deviate from the selected model (the straight model in the example below).



RANSAC

- There are valid points (inliers) that follow the selected model (the straight model in the example below).
- The goal is to find a straight line that minimizes the sum of squared distances.
- The invalid points (outliers), which do not follow the selected model, adversely affect the estimated model.



RANSAC

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Algorithm 15.4: RANSAC: fitting lines using random sample consensus

Determine:

- s** — the smallest number of points required
- N** — the number of iterations required
- d** — the threshold used to identify a point that fits well
- T** — the number of nearby points required
to assert a model fits well

Until **N** iterations have occurred

Draw a sample of **s** points from the data
uniformly and at random

Fit to that set of **s** points

For each data point outside the sample

Test the distance from the point to the line
against **d** if the distance from the point to the line
is less than **d** the point is close

end

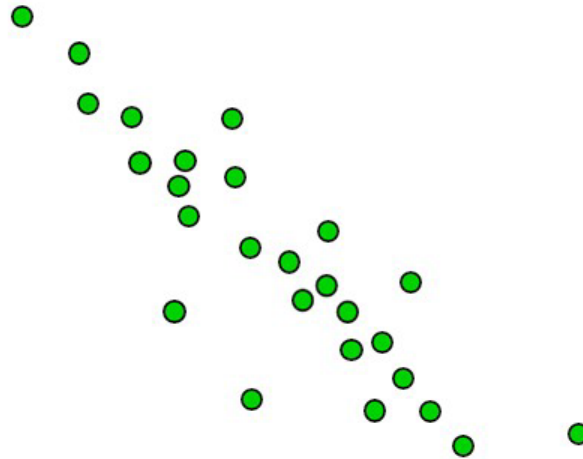
If there are **T** or more points close to the line
then there is a good fit. Refit the line using all
these points. Add the result to a collection of good fits.

end

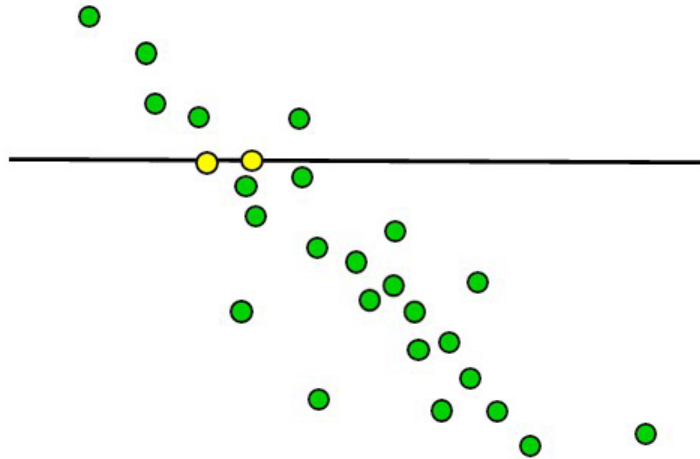
Use the best fit from this collection, using the
fitting error as a criterion

(Forsyth & Ponce) Slightly modified

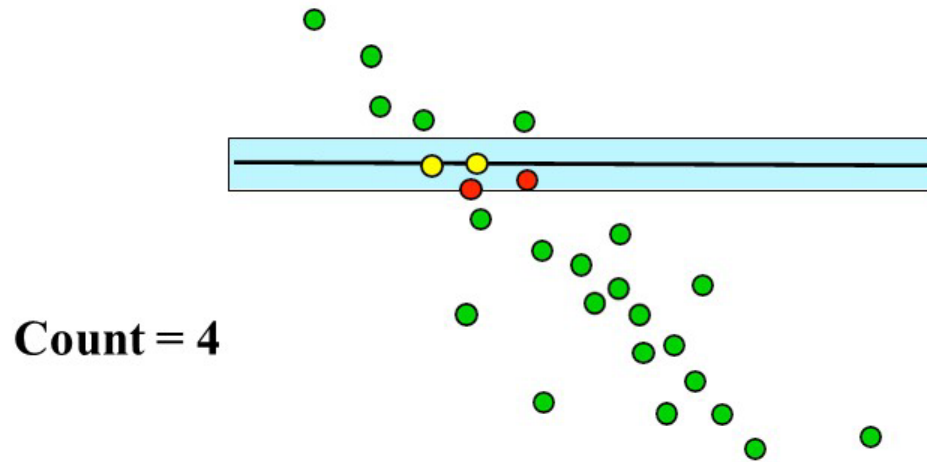
RANSAC



RANSAC

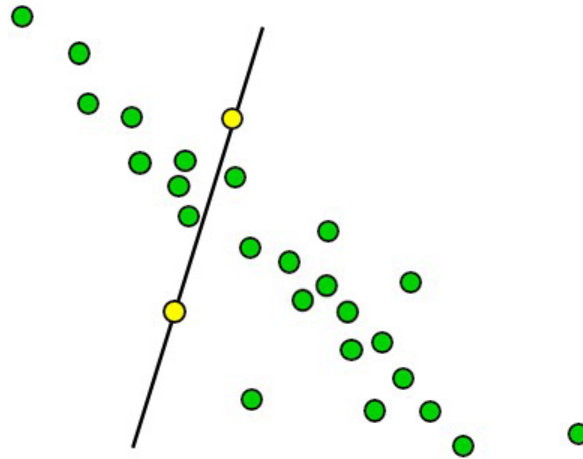


RANSAC

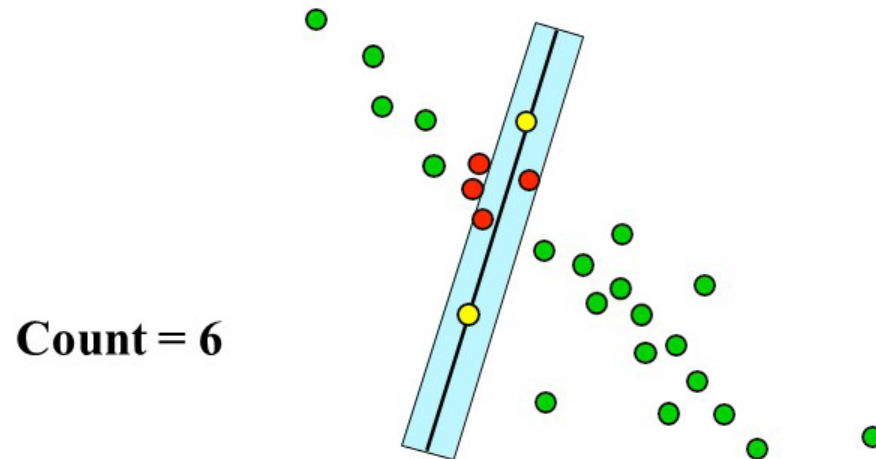


Count = The support for this line = number of points that lie within a distance threshold or the number of points that fit the model.

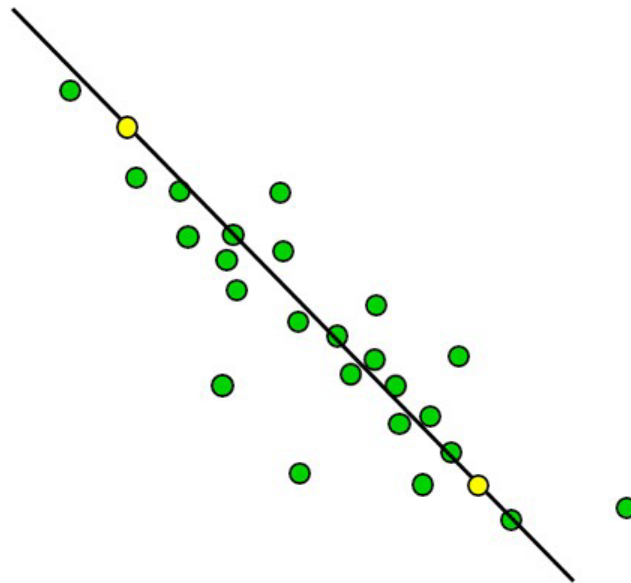
RANSAC



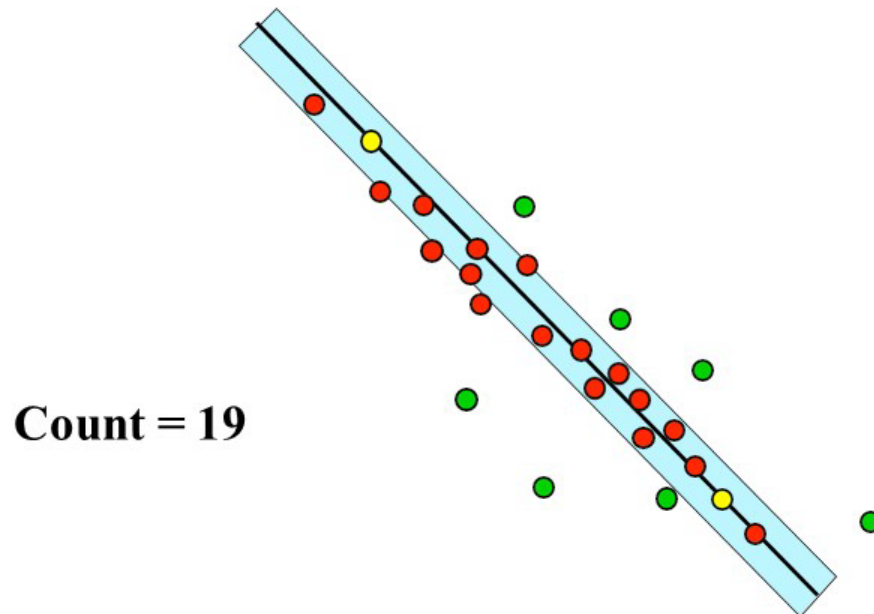
RANSAC



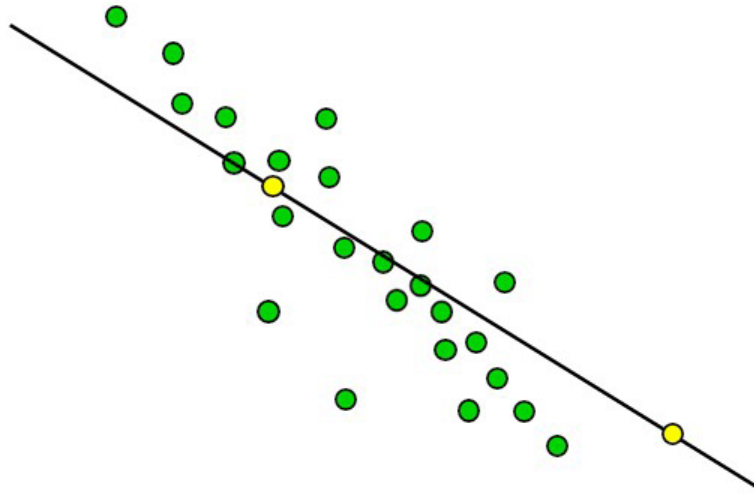
RANSAC



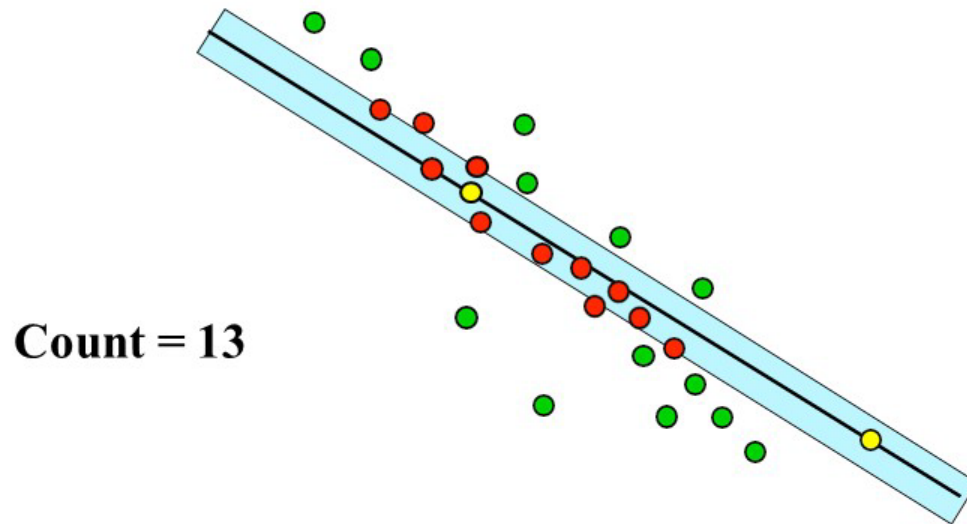
RANSAC



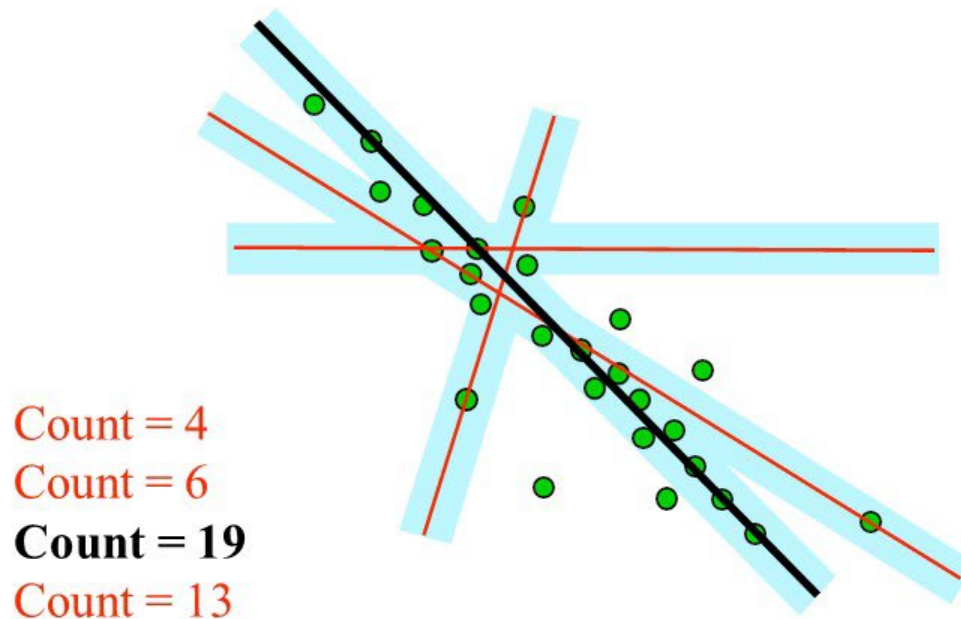
RANSAC



RANSAC



RANSAC



Count = The support for this line = number of points that lie within a distance threshold or the number of points that fit the model.

RANSAC

Choose N so that, with probability p , at least one random sample is free from outliers. e.g. $p=0.99$

$$(1 - (1 - e)^s)^N = 1 - p$$

$$N = \frac{\log(1 - p)}{\log(1 - (1 - e)^s)}$$

proportion of outliers e							
s	5%	10%	20%	25%	30%	40%	50%
2	2	3	5	6	7	11	17
3	3	4	7	9	11	19	35
4	3	5	9	13	17	34	72
5	4	6	12	17	26	57	146
6	4	7	16	24	37	97	293
7	4	8	20	33	54	163	588
8	5	9	26	44	78	272	1177